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REVIEW PAPER

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The influence of perceived risk on purchase intention in e-commerce—Systematic review and research agenda

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Abstract

This study systematically reviews the current state of research on the impact of perceived risk on purchase intention (PR&PI) in e-commerce to propose an agenda for future research. The authors selected and reviewed 133 studies conducted over the last decade. The theory-context-method (TCM) framework sheds light on developing trends in theories, contextual coverage, and methodological approaches in PR&PI research. Simultaneously, the stimulus-organism-response (SOR) framework reveals and categorizes the factors influencing perceived risk and purchase intention. The study enriches the current literature by proposing a research agenda to unpack gaps in the current literature and instruct future studies. Additionally, it diversifies the framework-based systematic review by applying the SOR framework with seven sectors visualized in a Venn diagram. This enhances the explanatory power of different types of variables and mechanisms of patterns in PR&PI research. Further, the review also inspires businesses to develop risk-reducing strategies and software. The study also provides e-retailers and e-sellers with an understanding of factors affecting consumers' risk perceptions, emotions and responses to e-commerce, allowing them to improve their performance.

KEYWORDS
e-commerce, perceived risk, purchase intention, SPAR-4-SLR, TCM framework, the SOR framework

1 | INTRODUCTION

Although the foundation of online commerce is trust, perceived risk is still considered an important indicator of behavioural intent. Bauer (2001) was one of the first scholars to introduce the concept of risk in marketing and since then, perceived risk theory has become the most widespread theory of human choice. Perceived risk is defined as 'the consumer's perception of the uncertainty and adverse consequences of buying products or services' (Dowling & Staelin, 1994, p. 114). With the appearance of the Internet and e-commerce, perceived risk refers to 'the potential for loss in the pursuit of a desired outcome of using an e-service' (Featherman & Pavlou, 2003, p. 454), or 'a consumer's expectation that the actions entailed in purchasing goods or a service from an e-commerce site could have unwanted outcomes' (Glover & Benbasat, 2010).

Despite a growing literature on perceived risk, current research is somewhat fragmented. This makes it difficult to draw meaningful and precise conclusions. For instance, literature reviews conducted before 2000, such as the one by Mitchell (1999) and Ross (1975), primarily focused on the context of physical shopping and its related concerns.

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Researchers have not thoroughly explored the online context or the factors driving perceived risk. The study by Maziriri and Chuchu (2017) failed to construct a systematic review of perceived risk and provided an insufficient system of factors affecting social risk. Pelaez et al. (2019) only investigated the relationship between perceived risk and purchase intention within the theories widely used, such as the technology acceptance model (TAM), the theory of planned behaviour (TPB), and the theory of reasoned action (TRA). From another perspective, the study by Rehman et al. (2020) reviewed the literature on perceived risk based on the model of antecedents, reducers, and consequents. However, the study omitted antecedent factors related to website functions from its analysis. Further, the factor of culture was categorized as the miscellaneous antecedent that impacts consumers' perceived risk.

Therefore, a systematic review of the influence of perceived risk on purchase intention (PR&PI) has been conducted to synthesize academic contributions and propose a research agenda. The study differs from existing review papers in some major ways:

This study integrates a bibliometric review and a framework-based systematic review to provide a throughout summary of PR&PI literature. A thematic cluster analysis is used as the background to assess and synthesize the findings into the theories, contexts, and methods (TCM) framework (Paul et al., 2017) and the stimulus-organism-response (SOR) paradigm (Mehrabian & Russell, 1980). This combination is required, as each framework alone is insufficient to explain aspects of the relationship between perceived risk and purchase intention in e-commerce. The SOR framework identifies and categorizes the factors surrounding the relationship between perceived risk and purchase intention in e-commerce. It also explains the dimensionality or relationships specific to the constructs (Jacoby, 2002). By contrast, the TCM framework explains the theoretical, methodological, and contextual drivers that underpin those constructs in PR&PI literature (Paul et al., 2017). Consequently, these two frameworks encompass the entirety of the current PR&PI literature regarding 'what we know and how we know it'.

The SOR paradigm is used to organize the findings in this study, as this framework has been frequently adopted to explain consumer behaviours in the literature (Zhu et al., 2020). Further, the impact of PR&PI in e-commerce is an internal, psychological process of consumers. Therefore, the SOR model is suitable for the explanation of a variety of phenomena, constructs and relationships that are both internal and external to the organism (consumer).

Unlike the traditional SOR model, the SOR in this study is categorized into seven sectors and visualized in a Venn diagram with overlapping areas, according to Jacoby (2002). Very often, scholars find it difficult to determine whether certain constructs should belong to stimulus, organism or response areas in the SOR model. For instance, it is not always clear whether the factors rooted in the psychographic and demographic characteristics of consumers are stimuli or organisms. Similarly, the experiential and memory factors, such as knowledge, experience and culture, of consumers should belong to stimuli or organisms. Therefore, this study contributes to the explanation of the role of variables in the PR&PI relationship. Further, the study develops an integrative conceptual framework with various patterns, such as the S-O, the S-R, the interplay and the S-O-R patterns. The conceptual model contributes to confirming that relationships between perceived risk, purchase intention and other influencing factors in the SOR model are sometimes psychological, fluid and dynamic.

In an attempt to understand the relationship between perceived risk and purchase intention in e-commerce, the authors expect the study to provide signposts for further research. It may offer references for practitioners, such as marketers and companies, to create risk-reduction strategies while stimulating consumers' purchase intentions in e-commerce.

2 | RESEARCH METHODOLOGY

The study applied the SPAR-4-SLR protocol introduced by Paul et al. (2021) to identify, select and synthesize existing literature. Figure 1 presents the process of assembling, arranging and assessing the studies.

2.1 | Assembling

During the assembling stage, the identification sub-stage identified the research domain and the following research questions:

- RQ1. What are the theories and models proposed to explain the effect of perceived risk on purchase intention in e-commerce?
- RQ2: What research context has examined the relationship between perceived risk and purchase intention in e-commerce?
- RQ3: What are the research methods adopted?
- RQ4: What are the approaches for modelling the concept of perceived risk?
- RQ5: What are the factors shaping the relationship between perceived risk and purchase intention in e-commerce?
- RQ6: What constitutes a synopsis of perceived risk and purchase intention patterns?
- RQ7. What are the research gaps and directions for future research?

To address these questions, the study adopted a hybrid SLR review, including bibliometric and framework-based review methods to facilitate critical discussion of themes through integrating relevant literature, synthesizing relevant studies, identifying knowledge gaps, and assisting future researchers (Dabić et al., 2020; Paul & Rowley, 2020). This study is based on analysing journal articles, as they contribute to academic progress and undergo a rigorous peer review process (Paul et al., 2021). Following Paul and Criado (2020), this study applied two widely used databases, Web of Science (WoS) and Scopus, to collect the list of potential studies. In addition, the authors also integrated Google Scholar to guarantee the inclusion of all pertinent articles.

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Assembling

Identification	
Domain:	Perceived risks and Purchase intention in E-commerce
Research questions:	RQ1. What are the theories and models proposed to explain the effect of Perceived risk on Purchase intention in e-commerce? RQ2: What is the research context that has been investigated in the relationship between Perceived risk and Purchase intention in e-commerce? RQ3: What are the research methods adopted? RQ4: What are the approaches to model concept of Perceived risk? RQ5: What are the factors shaping the relationship between Perceived risk and Purchase intention in e-commerce? RQ6: What constitutes a synopsis of perceived risk and purchase intention patterns? RQ7. What are the research gaps and directions for future research?
Source type:	Journal articles
Database:	Web of science (WoS), Scopus, Google scholar

Acquisition	
Keywords:	("Perceived risk") AND ("e-commerce" OR "electronic commerce" OR "online shopping") AND ("purchase intention" OR "buying intention" OR "willingness to buy" OR "purchase probability" OR "purchase likelihood" OR "purchase decision")
Search period	2010-2023
Initial screening criteria	• Peer-reviewed • SSCI/SCI-indexed (in WoS)
Total number of articles returned from the search	N=580 records WoS (n=174), Scopus (n=110), Google scholar (n=296)

Arranging

Organization	
Organizing codes:	Bibliometric review: Author's name, Title, Publication year, Source, Publisher, Author keywords, Keyword plus, Document type, Citation, Abstract, Language, DOI, Author's country TCM review: Theory & model, Research country, Population, Industry, Special context, Research approach, Statistical method, Measurement of perceived risk SOR review: Website factor, Marketing factor, Situational factor, Internal factor, Cognitive state, Affective state, Experimental storehouse (Long-term memory), response
Organizing Framework:	Combination of SOR (Mehrabian and Russell, 1980) and TCM (Paul and Singh, 2017) frameworks

Purification	
Article type exclude	Total article excluded (n=447): Duplicated paper (n=215), Inaccessibility or absence of the key topic (n=123), Theoretical research (n=12), Irrelevant journal or publisher (n=97)
Article type include	Articles included if a) being relevant to the domain, b) being in English, c) being published during 2010-2023, d) being accessible, e) Containing relevant keywords in title, abstract, and author's keywords, f) being empirical research, g) report research approach, statistical methods, sample N=133 records WoS (n=97), Scopus (n=17), Google scholar (n=20)

Assessing

Evaluation	
Analysis methods	Bibliometric modelling, descriptive analysis, thematic analysis
Agenda proposal method	Gap analysis

Reporting	
Reporting convention	Combination of discussion and summaries used. Limitation of the study are identified.

FIGURE 1 SPAR-4-SLR framework for systematic review (Paul et al., 2021).

In the acquisition sub-stage, a Boolean search using combinations of the keywords and their variations was run on these databases to include all relevant results, such as ('perceived risk') AND ('e-commerce' OR 'electronic commerce' OR 'online shopping') AND ('purchase intention' OR 'buying intention' OR 'willingness to buy' OR 'purchase probability' OR 'purchase likelihood' OR 'purchase

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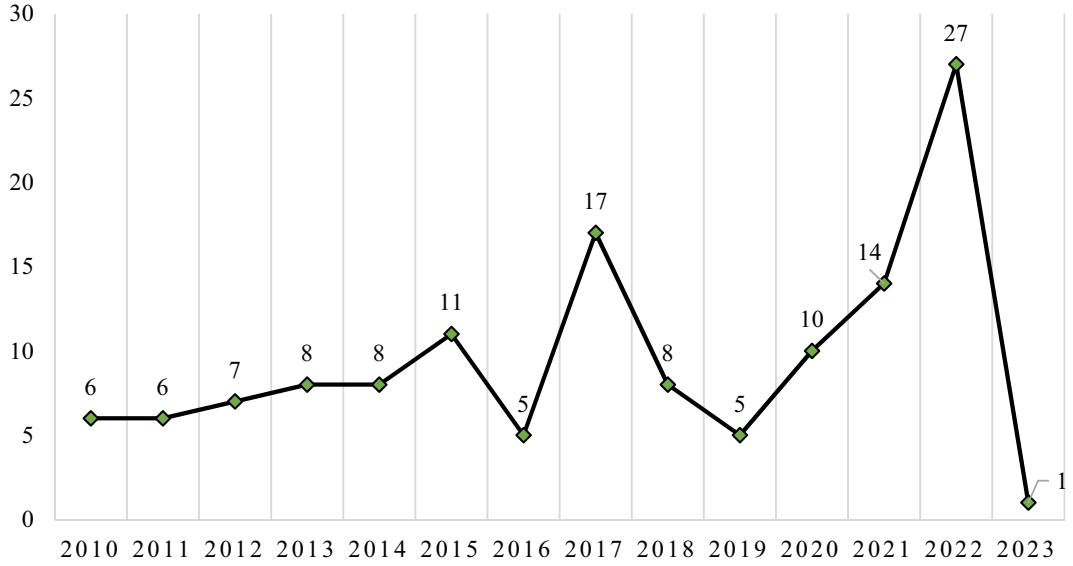


FIGURE 2 Annual scientific production.

decision'). The authors applied the 'Topic-All fields' query in the WoS database and the 'Tittle-Abstract-Keyword' query in the Scopus database. Google Scholar adopted the 'All fields' query. The study included academic articles published from 2010 to 2023, as research on the effect of PR&PI in e-commerce has increased rapidly during the current decade (Figure 2). The research findings also varied, ranging from having an insignificant effect to having a strong influence. Further, along with the exponential development of the Internet, e-commerce has also presented an unprecedented transformation since 2010, such as the variety of electronic payment tools (e.g., Google Wallet and Apple Pay), the introduction of social media shopping and mobile shopping. Additionally, security and privacy concerns also grew significantly with the growth of cybercrimes. For instance, Target suffered a massive data breach in 2013 (Jones, 2021) and eBay was hacked in 2014 (Reuters, 2014). Moreover, the appearance and spread of the COVID-19 pandemic at the end of 2019 and the surge in online shopping during lockdowns and social distancing measures profoundly impacted online consumer behaviour and risk perception. The end date of the electronic search was 25 March 2023. The initial search yielded a total of 580 records.

2.2 | Arrangement of data

The data were organized and purified in the second stage (Paul et al., 2021). In the sub-stage of organization of data, the articles were coded in terms of author's name, title, publication year, source, publisher, author keywords, keyword plus, document type, citation, abstract, language, DOI, author's country, theory & model, research country, population, industry, special context, research approach, statistical method, measurement of perceived risk, website factor, marketing factor, situational factor, internal factor, cognitive state, affective state, experiential storehouse (experiential and memorial factors) and response. These codes were combined and organized in line with the bibliometric analysis, the TCM and the SOR frameworks.

The authors performed a rigorous manual screening in the purification sub-stage using the excluded and included criteria to select high-quality and relevant studies. Of the total records, 215 studies were removed due to duplication; 365 studies were then assessed through reading titles, abstracts and keywords. The authors also removed 123 studies after screening because they were inaccessible and lacked a key topic. Twelve theoretical studies were also removed. Further, to ensure the credibility and reliability of research findings, articles' sources, including journals and publishers, were also examined. Consequently, the results removed 97 articles from controversial journals and publishers. Following the screening process, the qualitative assessment and synthesis included 133 eligible articles.

2.3 | Stage 3: Assessment

The third stage involved evaluation and reporting. The bibliometric modelling was performed by applying the bibliometric R package (Biblioshiny) to attain and document the quantitative data relating to the numerous articles selected (Aria & Cuccurullo, 2017). A content review performed both the descriptive and thematic analyses. Further, a gap analysis was also applied to analyse the research gaps and construct further research directions.

3 | BIBLIOMETRIC ANALYSIS

3.1 | Source analysis of articles investigated

Figure 3 presents the list of journals publishing more than one relevant study. It is observed that the study focused on certain journals

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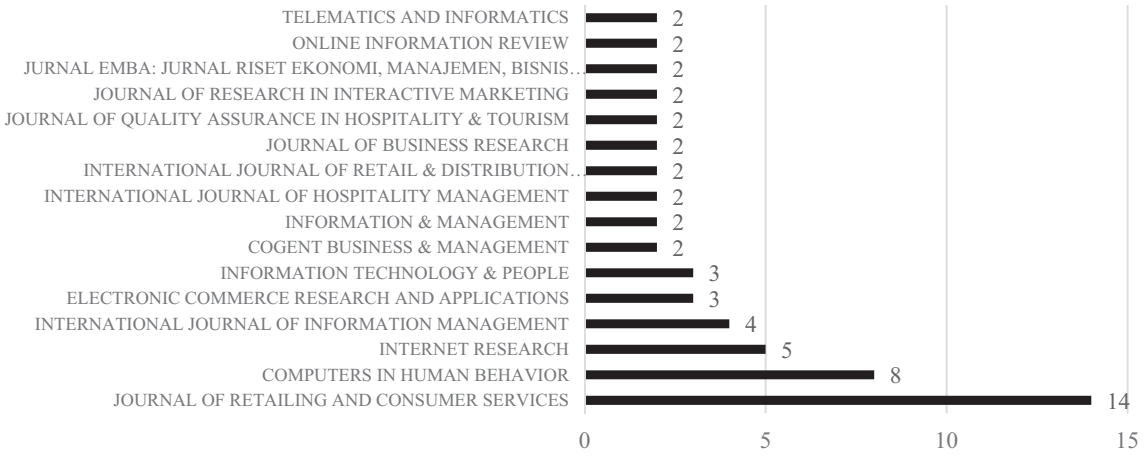


FIGURE 3 Sources with more than one relevant study.

that are dedicated to consumer behaviour on the Internet, such as the Journal of Retail and Consumer Services, Computers in Human Behaviour and Internet Research.

3.2 | Contributing authors' countries

Table 1 presents the list of countries' scientific production for the research domain. The results report single-country publications (SCPs) (i.e., the articles that are published by the researchers in the same country) and multiple-country publications (MCPs) (i.e., the articles that are published by the researchers from several countries). It is observed that the top five productive countries include China with 22 articles (SCP = 19, MCP = 18); the United States with 21 articles (SCP = 6, MCP = 15); Taiwan with 20 articles (SCP = 14, MCP = 6); Indonesia with 13 articles (SCP = 13, MCP = 0); and the United Kingdom with 11 articles (SCP = 3, MCP = 8). The United States has the highest MCP rate (MCP = 15), while Taiwan has the highest SCP rate (SCP = 6). Compared to the SCP rate, the MCP rate is higher in some countries, such as China, the United States and the United Kingdom. This outcome shows that these countries are more receptive to international cooperation in research than other countries.

3.3 | Authors' citation

There were over 9000 citation counts relating to the authors in this research domain. For instance, Chiu's publication from 2014 was highly quoted by other researchers. This figure is followed by Martins's publication in 2014, Kim's publication in 2013, Kesharwani's publication in 2012 and Amaro's publication in 2015 (Figure 4).

3.4 | Keyword co-occurrence and thematic map

VOSviewer has been used to visualize keywords and important clusters in the literature (Figure 5). The software facilitates the formation of clusters by classifying nodes in a network based on their interaction with words. Accordingly, all keywords were considered the unit of analysis, aided by the full counting method. A minimum of two occurrences of a keyword was set as a limiting factor. Consequently, 80 items were selected. The results show that perceived risk and purchase intention are among the most highly co-occurring keywords, with occurrence weights (total link strength) of 90 (248) and 42 (124), respectively. This implies their prominence over other items. Meanwhile, the distance of these keywords from other items indicates their relatedness and links. Further, three general clusters have been formed from the keywords. Cluster 1 (blue) represents commonly related terms to perceived risk, purchase intention and the factors associated with the product, brand and website. As such, the most common factors synthesized from this cluster include website quality (Chang et al., 2019; Park et al., 2012), seller reputation (Kim & Lennon, 2013; Park et al., 2012), product price (Pappas, 2016; Wu & Wang, 2011), and brand image (Shukla, 2014; Tam et al., 2024). Cluster 2 (red) focuses more on 'environmental/external' factors associated with community, society and culture. Accordingly, PR&PI are more interlinked with the keywords e-WOM (Amarullah et al., 2022; Ilhamalimy & Ali, 2021; Liao et al., 2021; Yuan et al., 2020), gender (Alrawad et al., 2023; Kasilingam, 2020), culture (Pratesi et al., 2021; Rosillo-Díaz et al., 2019) and social influence (Chopdar et al., 2018; Tandon et al., 2018). Cluster 3 (green) gives information on factors associated with individual perceptions and attitudes. As such, trust and attitude are the key points of the cluster.

Figure 6 describes the thematic cluster analysis of PR&PI research. The clustering was done by using the 'Walktrap' clustering algorithm. The research themes are characterized by two factors, including 'density', referring to the degree of development of a theme and 'centrality', indicating the significance of a theme in the growth of the study area (Cobo et al., 2011). Accordingly, the clusters were categorized into four quadrants.

Motor themes (the upper right quadrant) consist of themes such as TAM, TPB, security, price, quality, website design, website reputation, online trust, social influence and so on. The close interconnection of these topics results in a high level of density. They also have a

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TABLE 1 Country scientific production.

Country	Research articles	SCP	MCP
China	22	10	12
USA	21	6	15
Taiwan	20	14	6
Indonesia	13	13	0
UK	11	3	8
Malaysia	10	7	3
India	9	6	3
Spain	8	5	3
Canada	6	1	5
Korea	6	3	3
Pakistan	4	2	2
Portugal	4	3	1
Republic of Korea	4	0	1
Australia	3	2	1
Egypt	3	2	1
New Zealand	3	0	3
Saudi Arabia	3	0	3
Vietnam	3	3	0
Brazil	2	2	0
Italy	2	1	1
Jordan	2	1	1
Kuwait	2	0	2
Mexico	2	0	2
Qatar	2	0	1
Slovenia	2	1	1
South Africa	2	0	2
Thailand	2	2	0
Belgium	1	1	0
Bolivia	1	0	1
Ecuador	1	0	1
Fiji	1	1	0
France	1	1	0
Germany	1	0	1
Greece	1	0	1
Iran	1	0	1
Japan	1	0	1
Morocco	1	0	1
Netherland	1	0	1
Norway	1	0	1
Philippines	1	0	1
Romania	1	1	0
Sudan	1	0	1

Abbreviations: MCP, multiple-country publication; SCP, single-country publication.

direct connection to ideas used in other disciplines that have comparable themes, that is, they have a similarly significant level of importance. These topics are now propelling the advancement of the study area. The upper left quadrant encompasses themes such as green perceived quality and green perceived risk. These themes have low centrality but high density, indicating a greater concentration of internal connections and a lack of significant connections with other themes. These topics are considered niche themes with strong internal development but somewhat less influence on the overall study of PR&PI in e-commerce. The basic themes include perceived risk, purchase intention, trust, consumer behaviour, culture, structural equation modelling (SEM), e-WOM and so on. These topics have high centrality but have lower density values, indicating their significant conceptual connections with all other themes in the research area. However, they are not well-defined within their respective clusters. Although study areas consider these ideas essential, they require further internal development. The emerging themes at the bottom left corner have themes like information quality and mobile applications. These topics are developing due to their limited internal growth and restricted exterior connections with other themes. Research areas seem to be too nascent to have conceptual integration with other subjects. Apart from clear and distinct segmentations, some other topics are considered transitional and overlapping. For instance, the topics of COVID-19 and China are located between basic and motor themes, indicating the growing significance and maturity of these subjects throughout time. The topics of e-retailing, perceived enjoyment and subjective norms are categorized as overlapping with the basics and declining themes. This signifies that these topics are slowly acquiring the status of the basic topics in PR&PI research.

4 | DISTRIBUTION OF ARTICLES BY THE TCM FRAMEWORK

The TCM framework consists of three components: theories (T), contexts (C) and methods (M). Theories involve conceptual frameworks, models and theories examined by scholars for research purposes, while contexts include the specific situations involved in the examination, and procedures priorities the use of empirical facts to enhance the quality of the analysis. Paul and Singh (2017) stated that the TCM framework is advantageous for organizing the foundation of prior research. Additionally, it facilitates the discovery of novel insights by future researchers.

4.1 | Theories

RQ1. What are the theories proposed to explain the effect of perceived risk on purchase intention in e-commerce?

The results demonstrate that over 20 different theories, theoretical frameworks, and models are applied to explain the impact of PR&PI in e-commerce. However, only around 30% of studies reported the theories, frameworks, and models applied. It indicates that the current theoretical landscape of PR&PI in e-commerce may lack breadth and depth in scholarly insights. Furthermore, the results

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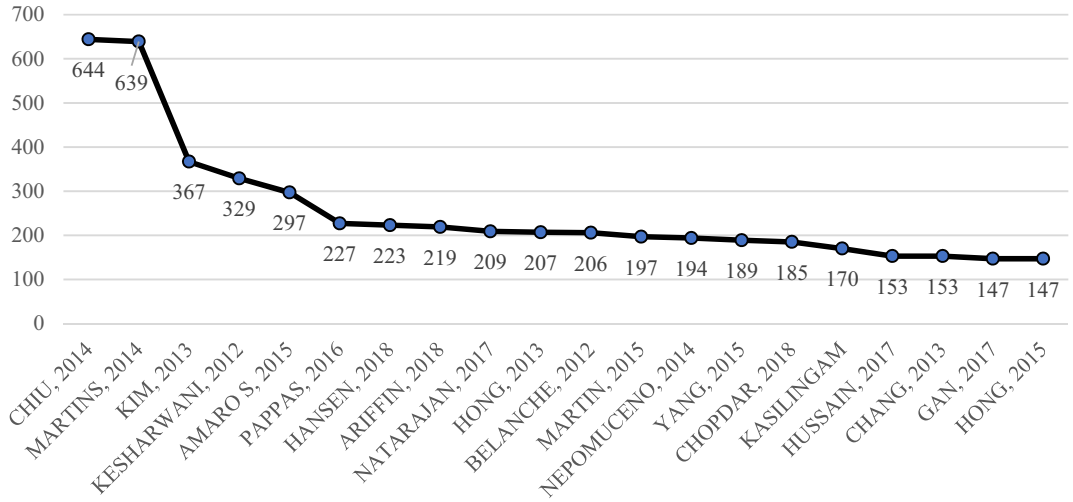


FIGURE 4 Top 20 documents with the highest citations.

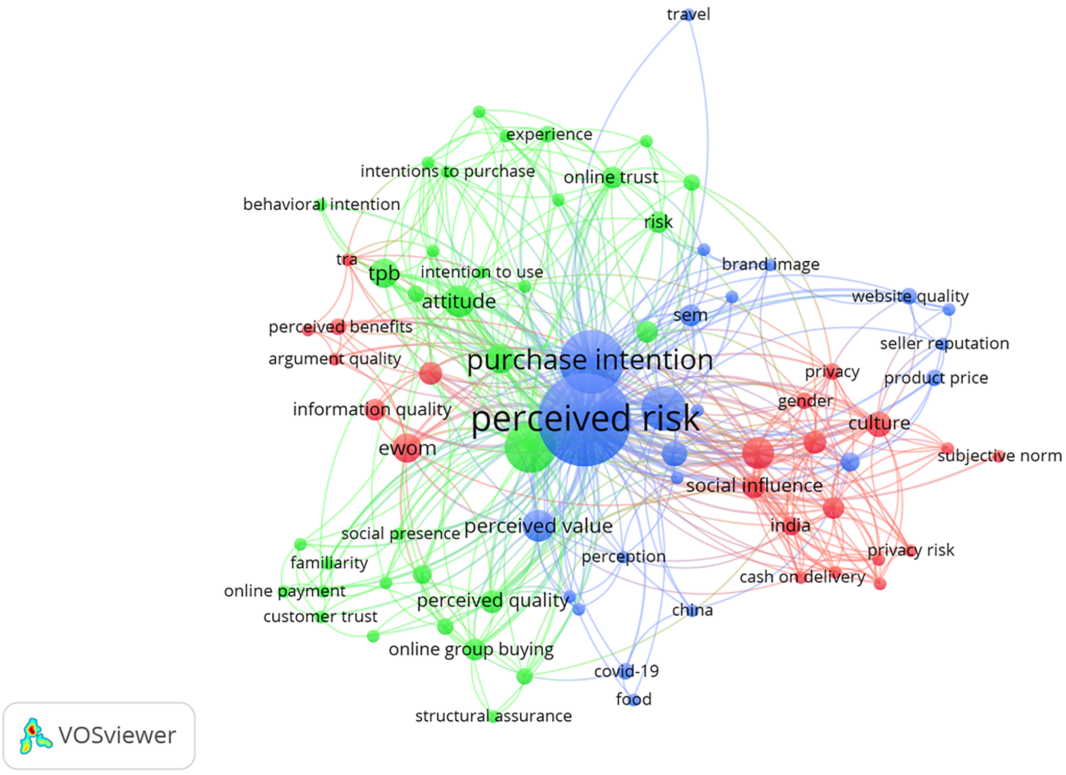


FIGURE 5 Co-occurrence map of the keywords.

found that 12% of studies adopted multiple theories or frameworks, indicating the potential for combining and deploying these theories in future PR&PI research. Table 2 presents theories, frameworks, and models applied in the literature. The majority of researchers applied the TAM and TPB in their studies. The theory and model are interrelated. TAM and TPB were derived from TRA (Fishbein & Ajzen, 1975) to explain consumer behaviour concerning new technology adoption. These theories are also the driving theories of PR&PI research. Further, several studies were found to apply the SOR framework to investigate the relationships between perceived risk and purchase intention in e-commerce. The SOR framework was introduced by Mehrabian and Russell (1980) to better reflect individuals' cognitive and affective states before their response behaviours in online environments.

4.2 | Contexts

RQ2: What research context has examined the relationship between perceived risk and purchase intention in e-commerce?

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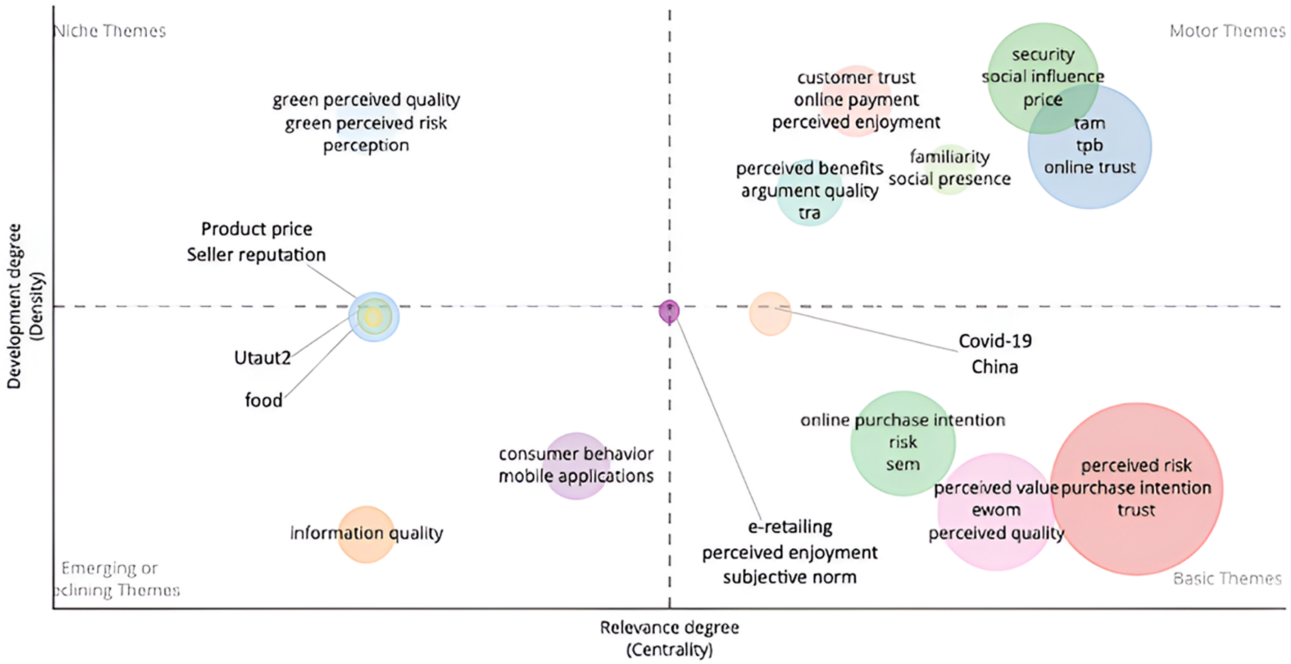


FIGURE 6 Thematic cluster analysis.

Paul et al. (2017) define context as the specific setting or circumstances in which the research took place. Table 3 offers a summary of contextual coverage. Accordingly, this review considered four main contexts: countries, respondents, industries and special contexts.

In terms of countries, 112 studies concentrate on one country. Twenty-one studies (15.7%) use samples that include more than one country. It implies that comparative research has been less dealt with. Further, researchers pay serious attention to Asian nations. Of these, roughly 18% (24 studies) focused on China as a context, with the largest e-commerce market in Asia. 11.27% (15 studies) focused on Taiwan, and 10% (14 studies) researched Indonesia. It is not surprising because they are also the nations with the highest production of PR&PI research. The United States market is also concerned with eight studies. Future research should scrutinize the underrepresentation of smaller markets like African, Central European and emerging markets, as their low representation is cause for concern.

In terms of respondents, e-shoppers are the most focused group of the samples, accounting for over 60% of total studies. Roughly 15.7% of the studies have used a sample of online users. The studies using a sample of students account for 13%. Further, only 9% of the studies used a discipline-specific sample (e.g., travellers, online banking users and social media users).

In terms of industries, most of the studies concentrate on a general context of e-commerce (97 studies), in which several studies also mentioned other e-commerce channels, such as mobile e-commerce and social e-commerce. Around 18% of the studies investigated a specific industry, such as the hospitality industry including tourism, food delivery services, and restaurant sectors; the food industry (e.g., the food distribution sector); and services (e.g., the banking and health care sectors). Further, roughly 9% of the studies focusing on the retail industry investigated the relationship between perceived risk and purchase intention on e-commerce platforms, including Jingdong, Taobao, Alibaba, Amazon, Aliexpress, eBay, Shopee and IKEA.

Significantly, the results also reveal some special contexts discussed in the literature, such as cultural differences and COVID-19 contexts. The review shows that only 7 out of 57 studies from 2019 to 2023 mentioned the context of COVID-19. Further, only 6 studies mentioned cultural differences as a special context, though 21 studies used a sample of more than one country.

4.3 | Methods

RQ3: What are the research methods adopted?

RQ4: What are the approaches to modelling the concept of perceived risk?

Methods include the various research methodologies and analytical techniques employed by scholars. The review only focuses on empirical research using primary data. The authors examined a total of 133 papers, taking into account research approaches and statistical techniques, to assess the methods described in the literature. Further, the methodological approaches to modelling the concept of perceived risk were also explored. Of these, over 96% of the studies applied a quantitative approach. Only three studies, accounting for 2.26% of the total, incorporated experiments into their research. Only one study combined a design of experiment (DOE) and a quantitative approach. Qualitative and experimental research approaches were the least used, which encourages future research to be pursued (Table 4).

Figure 7 shows the statistical methods adopted in the literature. It is observed that SEM is the most widely used method (116 studies).

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TABLE 2 Prominent theories, frameworks and models used in PR&PI research.

Theories/ frameworks/ models	No. of studies	Explanation	Exemplary papers
TAM	12	Ease of use and usefulness are the key factors of acceptance and use.	Tsai and Yeh (2010), Cheng et al. (2012), Kesharwani and Singh Bisht (2012), Nunkoo and Ramkissoon (2013), Amaro and Duarte (2015), Natarajan et al. (2017), Hansen et al. (2018), Han and Li (2020), Kasilingam and Lingam (2020), Wong et al. (2021); Carvache-Franco et al. (2022), Faqih (2022)
TPB	9	Behavioural attitudes, subjective norms and perceived behavioural control impact behavioural decisions.	Ali and Aziz (2022), Amaro and Duarte (2015), Bangun and Handra (2021), Hansen et al. (2018), Liao et al. (2010), Rahmi et al. (2022), Sinha and Singh (2017), Yu et al. (2018), Zhou and Liu (2022)
SOR framework	5	The stimuli exert an impact on the organism, often affecting the emotional, cognitive, or affective processes of a consumer, ultimately leading to their behavioural response.	Amarullah et al. (2022), Chang et al. (2019), Gong et al. (2023), He et al. (2022), Kim and Lennon (2013)
UTAUT	3	Performance Expectancy, Effort Expectancy, Social Influence and Facilitating Conditions are four core constructs that directly determine behaviour.	Chen et al. (2021), Chidambaram et al. (2024), Han and Li (2020)
Diffusion of innovation theory	2	Innovation as 'an idea, practice or object perceived as new by an individual or other unit of adoption'.	Kasilingam and Lingam (2020), Natarajan et al. (2017)
UTAUT2	2	Hedonic motivation, price value and habit are added the UTAUT.	Chopdar et al. (2018), Tandon et al. (2018)
TRA	2	Attitudes and subjective norms sharp behavioural intentions, which in turn determine behaviours	Amaro and Duarte (2015), Featherman et al. (2021)
Prospect theory	2	Individuals' behaviour is determined by their assessments of different options, which are influenced by both the potential consequences and their attitudes towards risk.	Chiu et al. (2014), Wong et al. (2021)
Signalling theory	2	The quality of a given cue is assessed based on its diagnostic and the presence of additional signals. In general, cues may be classified as information indices and information signals.	Meents and Verhagen (2018), Shao et al. (2021)
Equity theory	1	Individuals undergo motivational and cognitive processes by assessing the trade-offs or contributions (inputs) in relation to the benefits (outputs) and then contrasting the outcome with other comparable experiences or circumstances.	Lim (2020)
Risk-Benefit Framework	2	Consumers would consider benefits and risks simultaneously to make a final decision.	Featherman et al. (2021), He et al. (2022)
The protection-motivation theory	1	Individuals safeguard themselves by considering four key factors: perceived severity of a threatening event, perceived probability of occurrence or vulnerability, efficacy of recommended preventative behaviour, and perceived self-efficacy.	Gong et al. (2023)
Generation cohort theory	1	Age can impact consumers' attitudes, interests and shopping behaviour.	Sharma et al. (2022)
Social cognitive theory	1	Human behaviour is a value-driven behaviour.	Lho et al. (2022)
Value-based adoption theory	1	The perceived values and costs are considered as the antecedents to satisfaction and continuance intention.	Lho et al. (2022)
Two-factor theory	1	Individual' motivations are influenced by motivation factors and hygiene factors.	He et al. (2022)
Cue utilization theory	1	The use of a particular cue in evaluating quality differs depending on its diagnosticity and the presence of additional signals.	Shao et al. (2021)

(Continues)

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TABLE 2 (Continued)

Theories/ frameworks/ models	No. of studies	Explanation	Exemplary papers
Mean-end chain theory	1	Consumer behaviour is determined by the value they perceive in a product or service, which ultimately shapes their decision-making behaviours.	Chiu et al. (2014)
Flow theory	1	The individual is fully engrossed in an activity, completely unaware of themselves, which leads to a heightened sense of enjoyment.	Wu et al. (2020)
Social presence theory	1	The presence of the information sender influences recipients' understanding of the message.	Zhang et al. (2012)
Expectation-confirmation model	1	A customer's repurchase intention from their satisfaction in terms of a confirmation for expected and perceived performance of products/services.	Lho et al. (2022)
The CSLC framework	1	To develop a clear understanding of how information transparency is perceived and create a classification system.	Zhou et al. (2018)
Task-technology fit model	1	The capabilities of the technology characterized by task characteristics, technology characteristics, utilization and performance affect the individual's task.	Chen and Huang (2017)
The Affect Infusion Model	1	The influences of mood would become accelerated in complex and unanticipated situations.	Zhang et al. (2012)

Abbreviations: CSLC, The customer service life cycle framework; SOR, stimulus-organism-response; TAM, technology acceptance model; TPB, theory of planned behaviour; TRA, theory of reasoned action; UTAUT, The Unified Theory of Acceptance and Use of Technology.

Of these, 31 studies reported applying Partial least squares (PLS)-SEM to analyse the sample. Regression analyses are also preferred using several methods, such as multi-regression analysis and hierarchical regression analysis. Other methods, such as ANOVA, MANOVA and ANCOVA, were also applied. Further, two studies recorded adopted a mixed method.

Especially, the results demonstrate that most studies approached perceived risk as a uni-dimensional construct (92 studies) (Figure 8). On the other hand, 41 studies examined perceived risk as a multi-dimensional construct with several types of risk. In terms of methodology, the review explored two popular methods of modelling perceived risk, including lower-order and higher-order models. Of these, lower-order modelling is the most widely used, with 121 studies. Only 12 studies modelled and assessed the perceived risk as the higher-order variable.

5 | DISTRIBUTION OF ARTICLES BY THE SOR FRAMEWORK

RQ5: What are the factors shaping the relationship between Perceived risk and Purchase intention in e-commerce?

The theory analysis reveals a preference for the SOR framework to model and explain the relationship between perceived risk and purchase intention in e-commerce. Consequently, this review also applied this framework to organize the research findings. However, unlike other research that constructed a sequential SOR, this review visualises the SOR as a Venn diagram, and the research findings are categorized into encounter environments, automatic processing, experiential storehouse, consciousness and internal responses sectors (Figure 9). Jacoby (2002) suggests that certain groups of factors, like beliefs, attitudes, intentions, or satisfaction, exhibit overlapping or confluence areas instead of distinct segmentation. Therefore, a Venn diagram may offer better visualization.

5.1 | Sector 1. Encounter environment—External stimuli

Sector 1 consists of the external stimuli that the consumer or individual encounters at the moment. It includes three groups of factors such as (i) website-related factors, (ii) marketing-related factors and (iii) situational factors.

5.1.1 | Website stimuli

Website-related stimuli are the visual and auditory cues presented to consumers while browsing a shopping website. Numerous factors found in the literature have led to the frequent investigation of website-related stimuli. The findings of these investigations show that there are two primary groups of website-related stimuli, including website functionality and website quality. The factors associated with website functionality include website technology, website personalization, navigation, accessibility, vividness and so on, while the factors associated with website quality include security, privacy, website

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credibility, and information quality. Using surveys, most studies identified that a high-quality website or platform with effective functions mitigates consumers' perceived risk while increasing purchase intention in e-commerce. For instance, Kim and Lennon (2008) investigated the effect of website design, website reliability, service and security on perceived risk and purchase intention in e-commerce. Chang et al. (2019) and Cho et al. (2014) examine how website quality influences consumers' perceived risk. Generally, the relationships between website quality-related factors and perceived risk were studied more frequently, for instance, in the research of Bebber et al. (2017), Chen and Huang (2017), Soto-Acosta et al. (2014) and Wang (2017).

TABLE 3 Contextual coverage.

Context	No. of articles	% of articles
Countries		
China	24	18.045
Taiwan	15	11.278
Indonesia	14	10.526
India	8	6.015
United States	8	6.015
Malaysia	6	4.511
South Korea	5	3.759
Portugal	4	3.008
UK	4	3.008
Australia	2	1.504
Brazil	2	1.504
Egypt	2	1.504
Thailand	2	1.504
Vietnam	2	1.504
Bolivia	1	0.75
Canada	1	0.75
Chile	1	0.75
Germany	1	0.75
Korea	1	0.75
Kuwait	1	0.75
Mexico	1	0.75
Morocco	1	0.75
Netherland	1	0.75
Pakistan	1	0.75
Romania	1	0.75
Slovenia	1	0.75
South Africa	1	0.75
Spain	1	0.75
	112[a]	84.21%
Respondents		
E-shopper	83	62.406
Online banking user	1	0.752
Online user	21	15.789
Student	18	13.53
Traveler	6	4.51
Social media user	4	3.01
	133	100%
Industries		
Hospitality industry	15	11.28
Food industry	6	4.51
Retail industry	12	9.02
Service industry	3	2.26
	36[b]	27.07%
Special context		
COVID-19	7[c]	
Cultural difference context	6	

[a]Twenty-one papers used samples wherein more than one country is included.
[b]Ninety-seven studies used E-commerce general context.
[c]Counted in 57 studies from 2019 to 2023.

TABLE 4 Research approaches.

Research approach	No. of articles	% of articles
Primary data ($N = 133$)		
Quantitative	129	96.99
Experimental	3	2.26
Experimental and quantitative	1	0.75

5.1.2 | Marketing stimuli

Marketing stimuli include cues associated with brands, sellers and product factors. Generally, researchers examine factors related to the vendor or seller and the product more frequently than other factors. For instance, several studies investigated the impact of seller reputation on perceived risk and purchase intention. Other factors, such as vendor trustworthiness, vendor expertise, sellers' information, sellers' regulations (e.g., return policy and tariff refund), and the seller's capacity for transaction and logistics, were found to influence perceived risk and purchase intention. Factors relative to products, such as product price (Chopdar et al., 2018; Faqih, 2022; Pappas, 2016), product information (Meents & Verhagen, 2018; Tsai & Yeh, 2010), and product quality (Pappas, 2016), have also been concerned.

5.1.3 | Situational stimuli

Situational stimuli are associated with the society or environment that consumers live in. These stimuli, in particular, stem from online communities or legal aspects. Situational stimuli may increase or decrease consumers' perceived risk and purchase intention. e-WOM, or electronic

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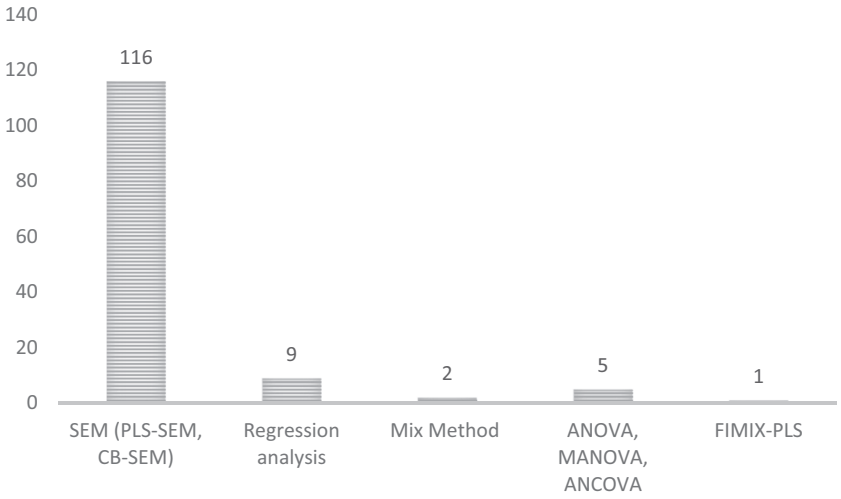


FIGURE 7 Statistical methods.

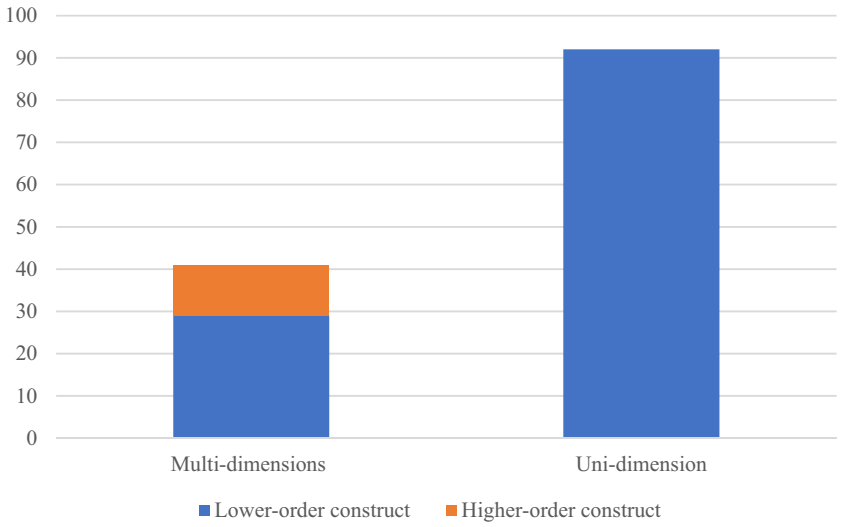


FIGURE 8 Modelling construct of perceived risk.

word-of-mouth, is a familiar stimulus that impacts consumers' risk perception (Liao et al., 2021; Lim, 2015; Nilashi, Abumalloh, et al., 2022; Nilashi, Ali Abumalloh, et al., 2022). Zhao et al. (2017) considered reference effects like social stimuli, as consumers' decisions are made based on the opinions of family and online communities. Similarly, He et al. (2022) also reveal that streamer expertise is also a social stimulus to lower perceived risk in e-commerce. Factors related to the legal aspect were least frequently investigated. Rouibah et al. (2016) showed that seals from a third party are considered lower perceived risk. Generally, community-related stimuli (e.g., e-WOM and social influence) have been studied more frequently, which indicates that in the online environment, consumers' perceived risk and purchase intention tend to be more affected by social factors.

5.2 | Sector 2: Automatic processing—Internal stimuli

Sector 2 is an overlapping area between stimulus and organism. Sector 2 involves internal stimuli that surface within the organism (Jacoby, 2002). These stimuli are non-visible. However, they can stimulate consumers' perceived risk and purchase intention. The review categorized these internal stimuli into demographic and psychographic characteristics (Satyavani & Chalam, 2018).

The results show that while psychographic factors were identified to directly impact consumers' perceived risk and purchase intention, demographic characteristics (e.g., gender, age and generation) often play moderating or controlling roles. Psychographic factors consisting of cognitive factors such as personal value (Mohseni et al., 2018), hedonic values and motivations (Chang & Tseng, 2013; Chiu et al., 2014; Pascual-Miguel et al., 2015) were found to affect consumers' risk perception and purchase intention. Further, inherent factors of the consumers (i.e., mood) also impact the perception of the risk (Faqih, 2022; Jordan et al., 2018; Zhang et al., 2012).

5.3 | Sector 3: Experiential storehouse

Experiential storehouse refers to the long-term memory of the consumer. It includes experiential and memorial factors associated with

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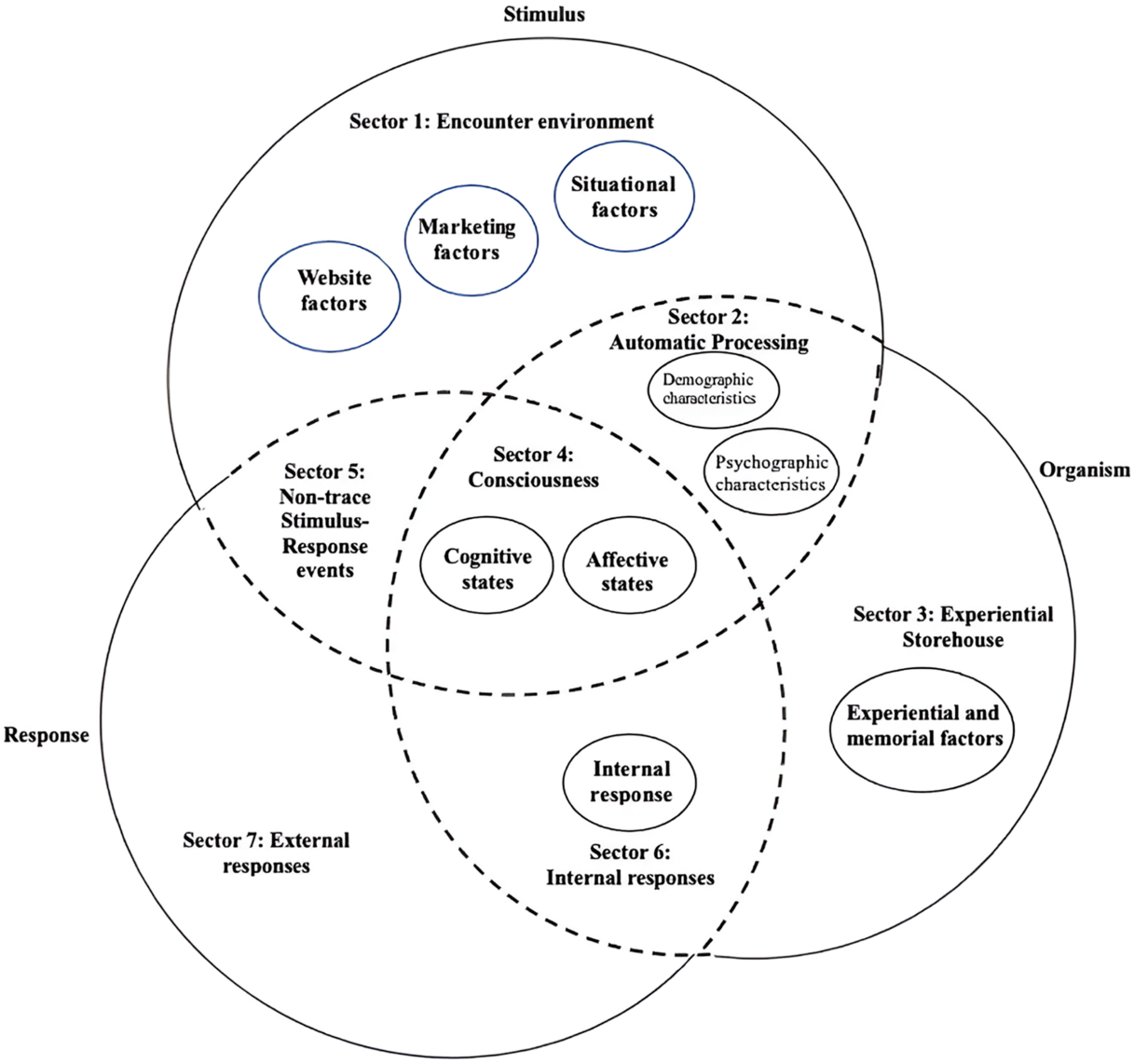


FIGURE 9 Visualization of the SOR framework. SOR, stimulus-organism-response.

prior experiences, knowledge, beliefs, awareness, and expectations, as well as the cultural context of the consumer. These factors are part of the environment encountered earlier. However, they were incorporated into the organism and introduced as a part of the organism. Several experiential storehouse factors were found in the literature, such as consumer expectation, experience, habit, familiarity, and knowledge. However, most studies identified the indirect effect of these factors on perceived risk via trust and satisfaction, such as those by Chopdar et al. (2018), Martin et al. (2015) and Zhu and Chih (2011).

The direct impact of consumers' experiential storehouses on PR&PI could be attributed to consumers' shopping experiences. This means that consumers with positive experiences in e-commerce perceive lower risks and tend to purchase on the platform (Martin et al., 2015; Mohseni et al., 2018). Rahmi et al. (2022) illustrate that brand awareness among consumers impacts perceived risk in e-commerce. Further, culture also resides in the organism as a factor of the experiential storehouse (Jacoby, 2002). It is defined as the collective programming of the mind that distinguishes the members of one category of people from others (Ganguly et al., 2010). Culture affects consumers' risk perceptions (Pratesi et al., 2021). Rosillo-Diaz et al. (2020) show that the perceived risk of e-commerce websites is more significant among consumers with higher uncertainty avoidance and among collectivist consumers.

Jacoby (2002) also reveals that many core phenomena across sectors can occur in the experiential storehouse. For instance, the organism receives information from external stimuli, which it then combines with information from the experiential storehouse to form new beliefs. This belief is then stored in the individual's experiential storehouse. For instance, the effect of consumers' trust on perceived risk has been different based on their experiences (Nilashi, Abumalloh, et al., 2022; Nilashi, Ali Abumalloh, et al., 2022). According to Park et al. (2012), cultural dimensions also moderate the relationships

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TABLE 5 Dimensions of perceived risk.

Types of perceived risk	Explanation	Study
Financial risks	The perception that a certain amount of money can be lost or is needed for a product to work correctly.	Alrawad et al. (2023), Ariff et al. (2014), Pillai et al. (2022), Rosillo-Díaz et al. (2019), Sharma and Crossler (2014)
Seller's fraud risk	Concerns about the reliability of online sellers, such as product information not reflecting actual quality, difficulty in finding a place to resolve disputes that arise when buying online.	Nguyen et al. (2021)
Performance risk	The possibility that consumers suffer stress due to their purchasing behaviours.	Hamed and El-Deeb (2020), Lin et al. (2022), Pentz et al. (2020), Sharma et al. (2022), Thakur and Srivastava (2015)
Time loss risk	Time lost due to difficulties navigating the website, submitting the product order and waiting for the product delivery.	Ariffin et al. (2018), Chen et al. (2012), Martin et al. (2015), Nepomuceno et al. (2014)
Social risk	Fear to be rejected or isolated by society because of the type of product they might buy or the online retailer that they might use.	Lho et al. (2022), Pentz et al. (2020), Sharma et al. (2022), Wu and Ke (2015)
Delivery risk	Lost or damaged products and shipping to the wrong location after buying.	Hong (2015), Rahmi et al. (2022), Yang et al. (2015)
Privacy risk	The probability for consumers feels angry due to online retailers use their personal information without permission and proper ways.	Chen and Huang (2017), Martin et al. (2015), Rahmi et al. (2022), Rosillo-Díaz et al. (2019), Sharma et al. (2022)
Psychological risk	A possibility for consumers scares self-image and self-concept will be lost and the products may not suit for their image due to the product purchased.	Alrawad et al. (2023), Ariff et al. (2014), Pillai et al. (2022), Rosillo-Díaz et al. (2019), Sharma and Crossler (2014)
Information risk	A possibility of losing control of personal information.	Alrawad et al. (2023), Nguyen et al. (2021)
Product risk	The perception that a purchased product may not perform as expected.	Ariffin et al. (2018), Pillai et al. (2022), Rahmi et al. (2022), Salim and Bahanan (2022)
Security risk	Unfavourable factors contributing to the slow growth rate of user acceptance of e-commerce.	Ariffin et al. (2018), Han and Kim (2017), Tandon et al. (2018)

between external stimuli and consumers' online trust, between online trust and perceived risk and willingness to purchase.

5.4 | Sector 4: Consciousness

Section 4 presents the confluence of stimulus, organism and response. The reason for this is to imply that humans typically do not sequentially analyse incoming inputs. Instead, while the organism receives and processes sensory inputs, it simultaneously responds (perhaps only internally) to those and other inputs. This sector includes the consumers' cognitive and affective states.

5.4.1 | Perceived risk as a cognitive state

Cognitive states pertain to the various concerns surrounding shoppers' interpretation of online information and the subsequent thoughts and beliefs regarding the product or service (Hajiheydari et al., 2017). Therefore, perceived risk can be one of the cognitive responses of customers during online shopping.

The review results show two basic research streams of perceived risk in e-commerce, including uni-dimension and multi-dimension research streams. The literature has investigated the uni-dimensional stream more frequently, with 92 studies. These studies mainly approach perceived risk as a single conceptual construct without identifying any specific risk types, such as those by Alimamy and Gnoth (2022), Chiu et al. (2014), Featherman and Pavlou (2003), Kesharwani and Singh Bisht (2012), Kim and Lennon (2013) and Malak et al. (2021).

In contrast, the multi-dimensional research stream proposes that understanding perceived risk is a multi-dimensional concept that encompasses various types of risk. Table 5 presents a compilation of prevalent sub-dimensions of perceived risk.

In addition, several studies also established multiple dimensions of perceived risk according to the specific conditions of their studies. For instance, Choi et al. (2013) categorized perceived risk in street food delivery into four sub-dimensions (i.e., hygienic risk, environmental risk, health risk and social-psychological risk). Bashir et al. (2021) provided a new categorization including 11 dimensions (i.e., high price risk, deception risk, transaction failure risk, dissimilar product risk, incapable service risk, illegitimate product risk, isolation risk, unease risk, displeasure risk, prior-purchase time delays risk and post-purchase time delays risk).

The selection of research streams in perceived risk studies could be attributed to the level of model complexity and the different

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impacts of perceived risk dimensions. For instance, Hong and Cha (2013) found that delivery risk insignificantly influences purchase intention. However, Hong (2015) showed that all risk dimensions significantly impact purchase intention.

5.4.2 | Affective state

Affective state involves emotional reactions arising when consumers interact with stimuli, long-term memories, and cognitive states. The review shows that trust, attitude and satisfaction have been examined frequently in the field. The concept of trust also varies in PR&PI research. Consumers' trust includes trust in brands (Gong et al., 2023), trust in sellers, trust in intermediaries (Malak et al., 2021), and trust in the Internet (Amaro & Duarte, 2015; Kesharwani & Singh Bisht, 2012; Yang et al., 2015). Further, most studies indicated that trust, attitude and satisfaction play a mediating role in the relationship between perceived risk and purchase intention in e-commerce. For instance, Hansen et al. (2018) found that perceived risk worsens consumers' attitudes towards e-commerce, lowering their intention to purchase. Perceived risk also contributes to lower consumer satisfaction and trust in e-commerce platforms. Consequently, consumers tend to refuse to purchase on sites associated with risk (Ilhamalimy & Ali, 2021; Kim & Koo, 2016; Martin et al., 2015).

The factor of emotion was underrepresented in PR&PI studies. Emotion is the set of clearly felt feelings that are triggered when using or consuming a product. It can be described by the different types of emotional experiences, like happiness, anger and fear (Kim & Lennon, 2013). The research of Kim and Lennon (2013) found that perceived risk generates negative emotions (fear and anxiety) for online consumers. Consequently, these negative emotions lower purchase intentions.

Furthermore, the review also illustrates that affective states may sometimes have an effect on perceived risk and even mutually influence each other. For instance, several studies investigating the relationship between trust and perceived risk found that trust diminishes consumers' risk perceptions in e-commerce (Amaro & Duarte, 2015; Kesharwani & Singh Bisht, 2012; Zhou et al., 2018).

5.5 | Sector 6. Internal response

Internal response refers to non-visible response. It is not directly visible to an outsider, and it results from prior conscious processing. The sector of internal responses is located in the confluence between organism and response, as internal responses are characterized by behavioural factors. However, they are invisible for the observer. Purchase intention is cited as an internal response variable in the literature and it is perceived as the willingness to engage in a purchase transaction (Hansen et al., 2018). A consumer's purchase intention is essential because it is the best predictor of consumers' actual behaviour. The study found that perceived risk negatively affects purchase intention, such as in studies by Liao et al. (2021), Mohseni et al. (2018), Sari et al. (2020), Sharma et al. (2022) and Vo et al. (2022). Additionally, purchase intention has also been explained through affective states, experiential factors of consumers and stimulus factors.

Figure 10 presents the overall findings from the TCM-SOR framework.

5.6 | A conceptual framework of PR&PI in E-commerce

Building on the SOR framework, a conceptual framework is proposed to offer a more comprehensive understanding of the existing knowledge by synthesizing the findings from the identified articles (Figure 11).

5.6.1 | Effect of stimulus factors on the organism (S-O pattern)

The S-O pattern has been categorized into two directions: stimulus-cognitive state and stimulus-affective state. Notably, encountering environmental stimuli does not guarantee cognitive or affective experience.

Supporting Information S1: Table A presents the direct effects of stimuli on perceived risk in e-commerce. Although most studies indicate a significantly negative relationship between these stimuli and perceived risk, some exceptions exist. For instance, Bebber et al. (2017) show that information quality and perceived risk do not correlate. Social media openness and social media speed were insignificantly influenced by the perceived risk (Permatasari & Kuswadi, 2017; Wang, 2017). Additionally, while channel transparency and channel uniformity negatively impact perceived risk, channel convenience insignificantly influences perceived risk (Xu & Jackson, 2019). Research findings for similar constructs have also revealed variations. For instance, e-WOM was found to mitigate consumers' perceived risk (Lim, 2015; Yuan et al., 2020). By contrast, Amaro and Duarte (2015) indicated that e-WOM (i.e., family and friend opinions) did not impact consumers' perceived risk.

Supporting Information S1: Table B reveals that most types of stimuli positively influence consumers' affective states (i.e., trust, attitude and satisfaction). For instance, the vividness and interactivity of the website improve consumers' attitudes towards purchase intentions (Lin et al., 2019). Kim and Lennon (2013) show that good seller reputation generates positive emotions in consumers.

5.6.2 | Effect of stimulus factors on response (S-R pattern)

The pattern of stimuli and responses shows that, in some cases, internal and external stimuli directly trigger consumers' purchase intentions in e-commerce. This is significant because the purchase

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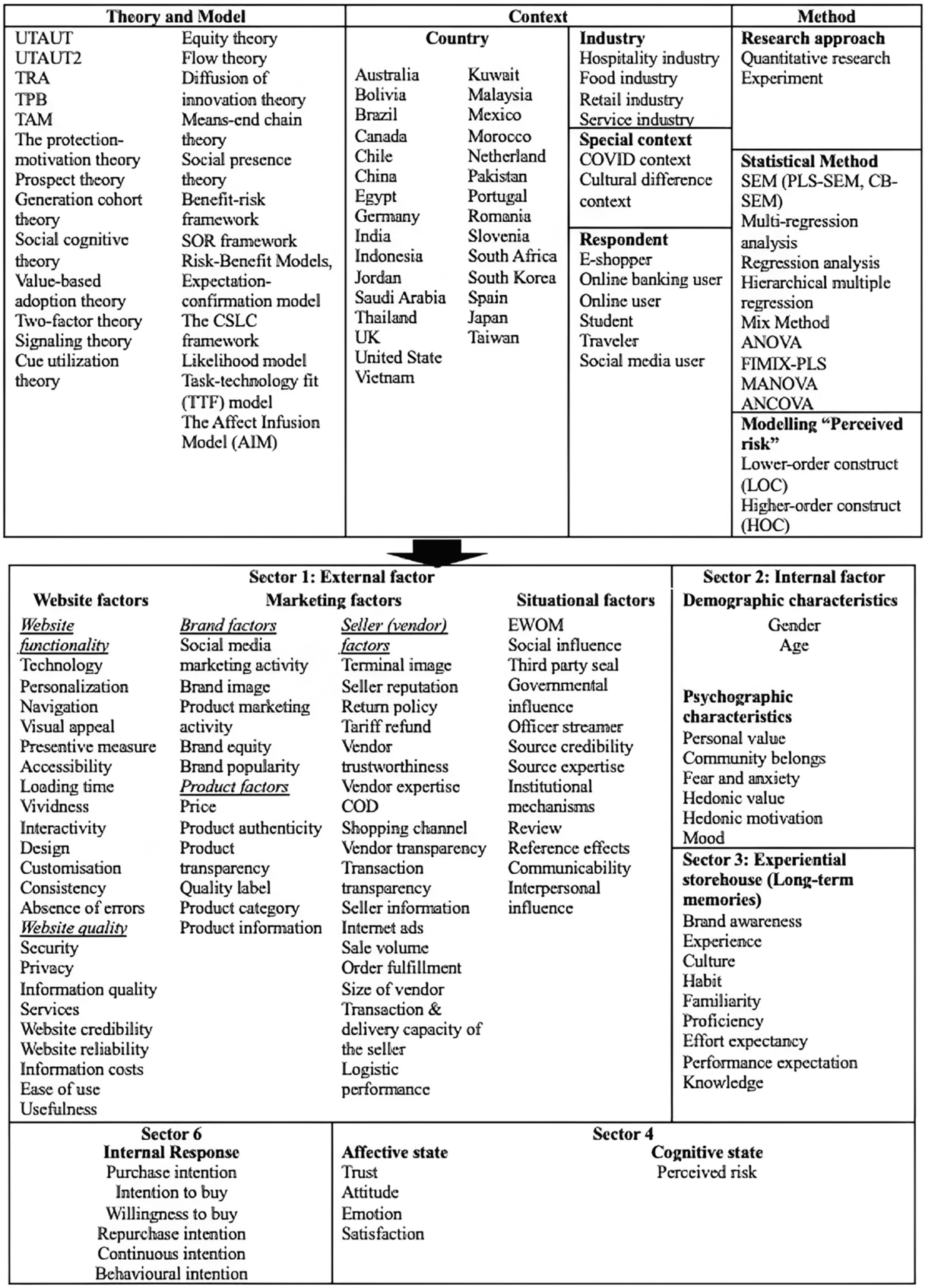


FIGURE 10 An integrative overview of findings from the TCM-SOR framework. TCM, theory-context-method; SOR, stimulus-organism-response.

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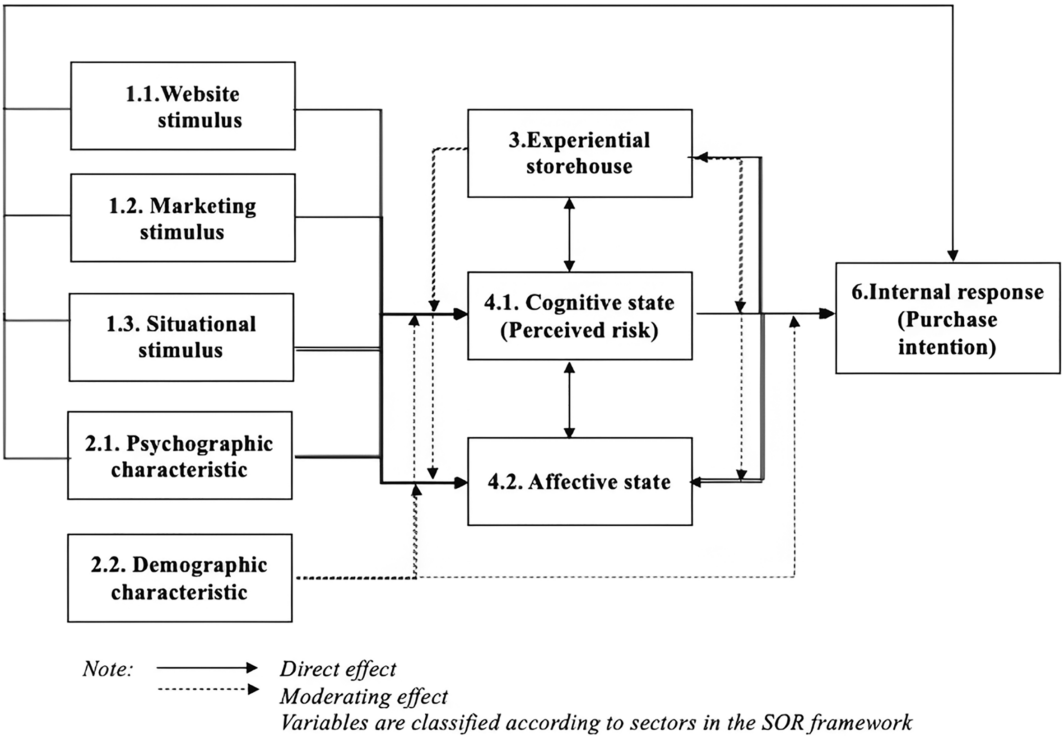


FIGURE 11 Conceptual model of PR&PI in e-commerce. PR&PI, perceived risk on purchase intention.

intention is regarded as an internal response, situated within the overlap between the organism and the response. Consequently, it may also be part of the organism. The review found that all four groups, including website, marketing, situational and psychographic factors, can directly affect purchase intention in e-commerce (Supporting Information S1: Table C). For instance, the personalization of the website, information quality and service directly promote consumers' purchase intentions in e-commerce (Alimamy & Gnoth, 2022; Soto-Acosta et al., 2014). Marketing factors such as online reviews, e-WOM, social influence, governmental influence and so on, also arouse consumers' intentions (Amarullah et al., 2022; Hsieh, 2015; Ventre & Kolbe, 2020). By contrast, other factors such as price, the website's security, and privacy have been recorded to insignificantly impact consumers' purchase intentions. It implies that, in addition to directly influencing the purchase intention, numerous stimuli must undergo multiple stages of processing before transforming into a response.

5.6.3 | The interplay between cognitive state, affective state and experiential storehouse

There are three types of interplay identified in the review (Supporting Information S1: Table D). First, several scholars found the relationship between cognitive state, affective state and experiential factors. For instance, Pratesi et al. (2021) found that consumers' culture can impact their perceived risk and trust in e-commerce. The shopping experience, knowledge and proficiency of consumers also influence their satisfaction, trust and risk perception (Nurunnisha, 2019; Sharma et al., 2022; Zhu & Chih, 2011). Second, most studies indicate that perceived risk has been considered the antecedent of affective states such as trust, attitude, emotion, and satisfaction. Accordingly, if consumers perceive a higher level of risk in e-commerce, they tend to decrease their trust, satisfaction, positive attitude and emotions. By contrast, the studies by Amaro and Duarte (2015) and Kesharwani and Singh Bisht (2012) reveal that trust can be the antecedent of perceived risk. If consumers trust the e-commerce website, online vendors and groups of reference members, their risk perception may be reduced (Hsu et al., 2014). Generally, the review implies that factors in the organism may not operate sequentially. Alternatively, the factors can inter-correlate and impact each other.

5.6.4 | Relationships between S-O-R pattern

The S-O-R relationship presents the sequential effects of stimulus factors on stages of the organism (i.e., cognitive state and affective state) and the internal response. The literature underrepresents the SOR theory and framework, despite the dominance of the S-O-R pattern. Supporting Information S1: Table E reveals two primary groups of S-O-R patterns. Researchers frequently examine a subset of the S-O-R pattern that excludes the cognitive or affective stages. For instance, Featherman et al. (2021) explored the impact of vendor expertise and trustworthiness on perceived risk and purchase intention in e-commerce. By contrast, other studies adopted a complete S-O-R model, such as Kim and Lennon (2013), Rouibah et al. (2016),

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TABLE 6 Future research questions.

	Gap area	Type of gap	Pertinent Research Questions (RQ)
Theory	In particular circumstances, perceived risk theory and cultural theory are not explained by the authors, although they are obvious from the discussion.	Theoretical	RQ1. Are perceived risk and cultural theories needed to further discuss?
	Theories applied to PR&PI are limited in TAM and TPB theories. Other behavioural economics, socio-psychological theories are less used.	Theoretical	RQ2: What behavioural economics and social-psychological theories can be applied to explain consumer perceived risk and purchase intention in e-commerce?
Context	Limited studies about the less-developed e-commerce markets	Population	RQ3: Do consumer's perceived risk and purchase intention vary in different countries or different cultures?
	Limited studies about the special context such as pandemic and world conflicts	Population	RQ4: Do consumer's perceived risk change in terms of force majeure context such as pandemic and world conflicts?
Methods	Need more qualitative and experimental studies	Methodology	RQ5: How can qualitative research and experimental research be conducted to explore consumer's PR&PI in e-commerce?
	Need more study in modelling perceived risk as a HOC	Methodology	RQ6: Should perceived risk be modelled as a higher-order construct?
Stimulus	Lack of studies on website functions in terms of checkout, customer support and cross-channel integration	Empirical	RQ7: What is the impact of checkout, customer support and cross-channel integration factor on PR&PI in e-commerce?
	Lack of studies on marketing and advertising stimuli	Empirical	RQ8: What is the impact of educational content and influences (KOC, KOL) on PR&PI in e-commerce?
	Limited studies on the effect of macro-situational stimuli	Empirical	RQ9: What is the impact of macro-situational stimuli on PR&PI in e-commerce?
	Limited studies on the psychographic characteristics	Empirical	RQ10: What is the impact of psychographic characteristics such as personality traits and lifestyles on PR&PI in e-commerce?
Organism	Limited studies on role of cultural factor	Empirical	RQ11: What is the role of cultural factors at the individual level in PR&PI in e-commerce?
	Limited studies on emotional factors	Empirical	RQ12: What is the role of negative feelings in the relationship between perceived risk and purchase intention in e-commerce?

Abbreviations: HOC, higher-order construct; KOC, key opinion customer; KOL, key opinion leader; PR&PI, perceived risk on purchase intention; TAM, technology acceptance model; TPB, theory of planned behaviour.

Shukla (2014), Sodadasi and Jyothi (2017) and Wu et al. (2015). Furthermore, the patterns, including S-cognitive state-R and S-affective state-R, were also investigated simultaneously. For instance, Amarullah et al. (2022) showed that e-WOM credibility impacts purchase intention via trust rather than perceived risk.

5.6.5 | Moderating effects

Moderating effects depict how stimuli and experiential factors may impact the strength or direction of the relationship between perceived risk and purchase intention in e-commerce. Supporting Information S1: Table F illustrates the moderating role of demographic characteristics (e.g., gender and age) in the effect of PR&PI. However, the results vary across studies. For instance, most studies found a significant moderating effect of age and gender (Chen et al., 2015; Natarajan et al., 2017). By contrast, the research of Alrawad et al. (2023) reports insignificant impacts of age and gender on the relationship between perceived risk and purchase intention. Further, external stimuli (e.g., website navigation), experiential factors (e.g., culture) and even affective states (e.g., trust and attitude) are sometimes modelled as moderators in the effect of perceived risk and purchase intention. It implies that the mechanism of factors affecting consumers' perceived risk and purchase intention in e-commerce may sometimes operate in time intervals rather than in time points. In other words, instead of an immediate and direct association, moderators gradually affect the conditions under which consumers' perceived risk influences their purchase intention.

6 | RECOMMENDATIONS FOR FUTURE RESEARCH

RQ7. What are the research gaps and directions for future research?

In this section, the study indicates various gaps that emerged from 133 studies (Table 6). Using these as a foundation, the study set

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a future research agenda associated with TCM and SOR. The study proposes several research questions and objectives to direct future investigations.

6.1 | Theory

The findings demonstrate that the authors did not explain perceived risk theory and cultural theory in particular circumstances although the discussion indicates that this should be obviously explained. Perceived risk theory is used to elucidate consumer behaviour in decision making. According to perceived risk theory, consumers are driven by the desire to minimize risk rather than maximize benefit when making a purchase (Mitchell, 1992). In general, researchers establish perceived risk according to the specific conditions of their studies. For instance, in the context of the hospitality industry, risk perception tends to be more pronounced when a product is novel or when there is little knowledge about it (Hwang & Choe, 2020). Similarly, as e-commerce gains popularity, perceived risk theory is also adopted to explain consumer's views about the possible negative consequences and uncertainties associated with participating in online transactions (Sharma et al., 2022). This calls for a discussion of the perceived risk theory with the behavioural intentions of e-commerce consumers at their core.

Further, the theme of culture has not been well defined in the PR&PI research yet, despite its significant conceptual connections. Cultural theory postulates that cultural factors shape several aspects of individual behaviour (Hofstede & Bond, 1988). Therefore, it can be applied to explain the variety of consumers' preferences and decisions (Brown et al., 2002; Chen et al., 2005). However, scholars failed to clarify whether their approach is national or individual. Using nations as units in the analysis of cultural differences is still controversial. Therefore, a deeper discussion of cultural theory is necessary to explain the differences between national and individual cultures. Further, other cultural models, such as GLOBE's or Trompenaar's theories (House et al., 2004; Smith et al., 1996), may also be of value. This indicates the direction of future research:

> D1. Future research should provide more explanation on perceived risk and cultural theories.

Second, that earlier, researchers have drawn heavily from theories such as TAM, TPB and TRA. These theories focus on explaining individual behaviours and behavioural intentions in the context of technology adoption, such as e-commerce. However, the study proposes that other theories and frameworks may be useful to unpack consumer behaviour from the perspectives of economics and social psychology.

Behavioural economic theories such as rational choice theory and game theory may be applied to explain consumer behaviour and risk perception. For instance, rational choice theory assumes that individuals make rational decisions based on their preferences and all relevant information. In social dimensions, it explains that user reviews and other factors can affect a group of people's reaction to a product, whether that reaction is favourable or unfavourable. This response can impact an individual's opinion and motivation to take action, such as purchasing. Furthermore, individual decisions to either buy or not buy a product can have a broader societal effect, resulting in a decrease or increase in the product's sales rank (Sajedikhah et al., 2024). The rational choice theory is meaningful in investigating the effect of perceived risk on consumer behaviour, as consumers are considered to analyse risk and benefit simultaneously before arriving at final decisions. A case in point is the study of He et al. (2022), who revealed that live streaming exposes online consumers to both benefits and perceived risks, such as product quality risk, personal information leakage and financial loss. Consequently, consumers' purchase intentions should be considered based on other factors relative to streamers, products or government credibility. As an extension of rational choice theory, game theory also has potential in this research domain. The basic principle of game theory is that the players make decisions based on their gained information (Rajbhandari & Snekkenes, 2011). However, the e-commerce environment is characterized by the information asymmetry in terms of product and service qualities existing between the buyers and sellers (Shao et al., 2021), between different online sellers, and between the sellers and the platform managers (Nana et al., 2022). Consequently, game theory may contribute to explaining the opportunistic behaviours existing in online transactions (PhamThi, 2022).

Psychological theories (e.g., prospect theory, hierarchy of effect and flow theory) may be useful in explaining consumers' risk perception and behaviours in e-commerce. For instance, prospect theory introduces several concepts, including certainty effect (a preference for more minor specific outcomes over larger risky ones), the reflection effect (increased risk-aversion when limiting losses), framing effect and the isolation effect (risk preferences vary based on framing and sequential structure), and magnitude perception (change in 8–10 feels different than 56–58) (Ruggeri et al., 2020). According to Chiu et al. (2014), prospect theory is useful as it explains the asymmetries in consumer behaviour between perceived losses and gains in online shopping, elucidating the role of emotions and risk perception. Further, several studies mentioned the variety of genders and generations. For instance, Faqih (2016) found that gender plays a moderating role in the relationships between perceived risk and behavioural intention. In addition, consumers from different generations also deal with perceived risk and purchase intention differently (Sharma et al., 2022). However, there were limited studies that mentioned these theories in their discussions. Therefore, the review suggests there should be more social and psychological theories related to gender and age incorporated into research. For instance, gender schema theory is also meaningful to explain significant differences between male and female customers in purchase intention. This suggests that:

> D2. Future research should explore behavioural economics and social-psychological theories and models to explain consumers' perceived risk and purchase intention in e-commerce.

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6.2 | Context

The research on PR&PI in e-commerce has been predominantly conducted in highly developed e-commerce markets, such as China, the United States, India and so on, with less than a fourth of the studies in less developed and emerging e-commerce markets. According to Ventre and Kolbe (2020), perceived risk tends to be less for frequent online shoppers. Therefore, it might not be a barrier to the decision-making process of experienced online shoppers. By contrast, consumers with less experience making online purchases in emerging markets may perceive more risk than others (Thakur & Srivastava, 2015). Further, cultural differences can also be a meaningful context in this field, as consumers' behaviour and perceptions are not homogenous across nations and cultures (De Mooij, 2019). Consequently, comparative studies should be conducted.

Second, it may be interesting to see how perceived risk influences online consumer purchase intention in special contexts, such as pandemic and world conflict contexts. The COVID-19 pandemic has received less attention in the past. According to Nguyen et al. (2021), purchasing food online was booming during the COVID-19 pandemic in Vietnam due to governmental regulations such as social distancing. Therefore, the perception of risk may change when consumers have more experience with online purchasing. The context of COVID-19 was also an opportunity for marketing tools such as e-WOM to share knowledge and experience. Thus, it contributes to the impact on the relationships between perceived risk and purchase intention (Nilashi, Abumalloh, et al., 2022; Nilashi, Ali Abumalloh, et al., 2022). On the other hand, the outbreak of the pandemic resulted in a heavy dependence on online shopping. It raises major concerns about perceived privacy, security, and delivery risks (Gong et al., 2023). The results signal the following research direction:

> D3. Future research should investigate consumers' perceived risk and purchase intention in different countries, cultures, and special contexts.

6.3 | Methods

The results indicate that too much focus was given to quantitative research with an SEM analysis, which resulted in the replication of the research results. Quantitative research and SEM analysis are powerful tools for investigating complex models. However, most prior studies were limited to widely used methods and techniques, such as cross-sectional data collection and non-probability sampling techniques. Cross-sectional data collection was preferred in this field due to its convenience. However, this type of data collection only offers a snapshot of consumers' risk perception and purchase intention at a given point in time. Therefore, a longitudinal study may be more meaningful to assess the causality of the relationships and the change in consumers' risk perception. Additionally, it may be effective for studies focusing on the impact of stimulus factors on consumers' perceived risk and purchase intention. Further, survey and non-random sampling techniques are prevalent in this review. This approach may impact the representativeness of the whole population due to the likelihood of sampling bias. This has also been one of the challenges for researchers due to their lack of permission to access a list of the population. However, the work of Belanche et al. (2012) suggested an interesting technique to support the representative nature of the sample. The socio-demographic characteristics of the sample were compared to available information about a wider population. The similarity in terms of the age and gender of the respondents compared to the population can contribute to supporting the representativeness of the data. Additionally, it might be a promising endeavour to apply neurophysiological methods, such as electroencephalography (EEG) and electrodermal activity (EDA), to capture perceived risk and its emotional aspects instead of constructs using Likert scales. EDA can be utilized to record consumers' emotional variables, such as enjoyment, fear, nervousness, and anxiety (Erdos & Horvath, 2023). By contrast, EEG supports recording brain activities in risk processing (Baltruschat et al., 2021).

In addition to showing that studies mostly use quantitative research approach, the review reveals an underrepresentation of qualitative and experimental methods in this domain. The research of PR&PI requires qualitative research methodologies like focus groups and in-depth interviews, which enable researchers to investigate novel phenomena that impact the relationship between perceived risk and purchase intention. The qualitative method is also better at capturing the social context and cultural nuances that shape an individual's experience (Khan & Razzaque, 2023). In addition, the technique of netnography might be useful for studying online communities, cultures, and social interactions and understanding consumer perception and behaviour in an online environment (Salam et al., 2022). Further, the DOE is also encouraged in this field for studying the impact of stimulus factors, such as online reviews, website stimuli, and marketing stimuli, on consumers' perceived risk and purchase intention. Therefore, the results suggest that:

> D4. Qualitative and experimental research should be applied when exploring consumers' perceived risk and purchase intention in e-commerce.

Further, the review shows that the construct of perceived risk was primarily modelled as a lower-order construct (LOC), although the multidimensional approach has been popular in this research field. According to Koufteros et al. (2009), multi-dimensional constructs with a strong correlation may not provide a clear factor structure when submitted to factor analysis and hypothesis testing, whereas a higher-order model acknowledges the significance and preserves the unique characteristics of individual LOCs, seeing them as different aspects of the overarching higher-order construct. Therefore, constructing higher-order constructs may contribute to reducing model complexity and alleviating numerous methodological problems that challenge empirical studies. Therefore, the review suggests that

> D5. The construct of perceived risk should be modelled as a higher-order construct.

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6.4 | Stimulus

The literature has discussed a wide range of themes, but by focusing on aspects that are highly relevant to consumers, future researchers can further investigate the stimuli that impact PR&PI in e-commerce.

Initially, the review revealed a thorough investigation of the website's functional aspects, including website designs, user interfaces, loading speeds and quality-related factors. However, there is a lack of focus on the check-out and customer support functions on the e-commerce website. Consumers' purchase intentions are highly affected by their security and security risks (Chopdar et al., 2018; Tsai & Yeh, 2010). Therefore, checkout functions such as a simple checkout process and secure payment gateways may stimulate consumers' risk perception and purchase intention in e-commerce. Additionally, functions associated with website customer support may be interesting, such as live chat support or AI chatbots. Ali and Aziz (2022) touched on the aspect of website credibility and trustworthiness, but future research could potentially explore factors like trust badges and verified seals. Further, the aspect of mobile optimization on the website has been underrepresented. The uniformity of shopping channels lowers consumers' perceived risk (Xu & Jackson, 2019). Therefore, the website's responsive design for mobile devices and seamless integration can be further investigated.

Second, marketing stimuli investigated in the literature to a great degree related to the seller and product factors, such as quality, information, transparency and so on. The review suggests that future research may further investigate the impact of factors associated with free trials or samples (i.e., free trials and sample offerings) on consumers' risk perception and purchase intention. One of the major concerns of online consumers is product quality risk (Han & Kim, 2017). Therefore, stimuli relative to sample offerings may contribute to minimizing the perceived risk of product quality. It is also considered a marketing stimulus to minimize the seller's fraud risk. Further, the review also indicates that factors associated with marketing and advertising can have potential in this field. Social marketing, product market activities, and internet ads have been indicated to mitigate consumers' perceived risk while promoting willingness to purchase (Chafidon et al., 2022; Lim, 2015; Tam et al., 2024). However, scholars might further examine factors related to content marketing, specifically educational content and influencer marketing.

Regarding situational factors, factors such as e-WOM, review, reference effect and so on, have been frequently investigated. However, these factors should not be limited to a special shopping channel. Cross-channel communication is suggested to be a promising situational stimulus, as e-consumers can approach multiple sources of information during a purchase in e-commerce. This might decrease consumers' perceived risk while increasing purchase intention. Further, there is a lack of investigation of macro-situational factors in the broader environment. Although these factors are exogenous to the particular e-commerce business, they may impact how customers perceive the overall e-commerce environment. For instance, the stability of the economy might impact customer confidence and purchase behaviour (Sharma & Sonwalkar, 2013). Accordingly, economic downturns may lead consumers to perceive higher financial risk, resulting in a heightened examination of e-commerce purchases. Meanwhile, force majeure events, such as pandemics, geopolitical tensions or natural disasters, may be the reasons for a higher perceived risk related to the supply chain, logistics, and delivery.

Furthermore, exploring psychographic characteristics like personality traits and lifestyles is promising as personality traits can influence how consumers perceive and respond to risks in e-commerce (Wu & Ke, 2015). Therefore, personality traits such as risk aversion, conscientiousness, neuroticism, extroversion and so on, may be worthy of investigation. For instance, neuroticism and extroversion can impact consumers' perceived risk. Particularly, extrovert individuals may be less risk-averse, while neurotic individuals may tend to be more risk-averse (Oehler & Wedlich, 2018). The reason is that neurotic people display higher emotional reactivity, increased susceptibility to stress and a propensity for anxiety; thus, they tend to perceive more risk in online transactions. Likewise, the lifestyle of consumers may also impact their risk perception and purchase intention, as consumers focusing on health and well-being may perceive more risk in e-commerce. The thematic analysis also reveals that the topic of green perceived risk is highly developed but less relevant to the PR&PI field. Therefore, it may be further explored.

The review suggests that:

> D6. Future research can explore more stimuli in terms of website functionality, marketing, cross-channel communication, macro-situational factors, and socio-psychological characteristics.

6.5 | Organism

In contrast, an array of factors associated with organisms contributing to the relationship between perceived risk and purchase intention in e-commerce has not been investigated. For instance, the experiential factors residing somewhere in consumers' long-term memories can be further considered. The review shows that most studies investigated the relationships between factors and PR&PI in e-commerce at a cross-sectional point in time. As consumers change over time, their prior experience, awareness, knowledge and cultural values may integrate new stimuli to change their perceived risk and purchase intention (Jacoby, 2002). The thematic analysis also indicates that the topics related to the familiarity, experience and culture of consumers have a higher degree of development and relevance to the PR&PI field. Therefore, scholars should thoroughly investigate the role of experiential factors. In addition, the cultural factor has been underrated. Despite the mention of cultural differences and cross-cultural aspects, PR&PI studies lack a thorough understanding of cultural approaches. Therefore, the focus should be on the relationship between cultural variables and PR&PI, as varying cultural values could influence the perception of risks in e-commerce. Additionally, there has been widespread use of two distinct approaches to culture, namely the national and individual approaches. The application of the

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national approach reveals that national differences mirror cultural differences (Karahanna, 2013). However, the individual stream argues that national borders do not always correlate with culture. In some cases, individuals in society do not even have the same cultural values (Yoo et al., 2011). Consequently, a deeper understanding of the operation of culture may be meaningful.

Regarding the affective state, the emotional aspect has been under-represented, especially the impact of negative feelings on the relationship between perceived risk and purchase intention. According to Abdellaoui et al. (2007), negative emotional impacts are more significant than positive emotional ones. Therefore, negative feelings (e.g., worry, anxiety, threat and fear) may be a part of the decision-making journey. Future research can also incorporate the role of negative feelings and consumers' cultural values in PR&PI in e-commerce because consumers' negative feelings may sometimes derive from different cultural values.

The interplay between cognitive stage (i.e., perceived risk) and affective state should be further discussed due to the debate about whether cognition precedes emotion or emotion precedes cognition. This review observes these two perspectives. For instance, Kim and Lennon (2013) found that perceived risk is the antecedent of emotion. Conversely, research suggests that trust serves as a predictive factor for perceived risk (Malak et al., 2021; Nilashi, Abumalloh, et al., 2022; Nilashi, Ali Abumalloh, et al., 2022).

The review suggests that:

> D7: Future studies can further understand organisms by investigating the roles of consumers' experiential factors, such as culture and prior experience and affective states, such as negative feelings.

7 | IMPLICATIONS

7.1 | Theoretical implications

By integrating a bibliometric analysis and a framework-based review of the TCM and SOR models, the review offers several theoretical insights for this field. First, the review conducted a comprehensive analysis of the literature using the TCM perspective. This analysis identified developing trends in theories, contextual variables, and methodologies in PR&PI research. Accordingly, the review reveals the central role of theories such as TAM, TPB, and TRA in explaining consumers' perceived risk and purchase intention in e-commerce. It also reveals what are the frequent methodologies (e.g., SEM, survey and lower-order modelling) adopted by the researchers in this field. Consequently, the TCM framework facilitates the review to unpack promising theories, contexts, and methods in PR&PI research.

Second, the review consolidates factors associated with PR&PI in e-commerce into the SOR framework. This generates several positive features. Instead of a traditional fundamental model, this SOR framework is visualized in a Venn diagram with seven sectors. This implies that, apart from being simultaneously compatible with the sequential SOR model with three primary factors, the model accommodates controversial factors by locating them in overlapping areas. The authors classify stimulus factors into two categories: external factors, which refer to encountering environmental factors like websites and marketing, and internal factors, which refer to the demographic and psychographic characteristics of consumers. Internal stimuli are categorized between the stimulus and organism areas, as they are rooted in the consumer but able to arouse consumers' perceived risk and purchase intention. The organism is categorized into the cognitive state (i.e., risk perception), affective state (i.e., consumers' emotion, trust, satisfaction and attitude), and experiential and memory factors (e.g., consumers' culture, experience and knowledge). The cognitive and affective states are located in the confluence of stimulus, organism, and response. This implies that consumers may not receive and process sensory inputs sequentially. Finally, purchase intention is segmented in the internal response sector (i.e., between the organism and the response), as it is an invisible behavioural factor. This framework is believed to be exhaustive in identifying types of variables and understanding mechanisms of patterns in PR&PI research, as it makes it easier to explain certain cross-sector and dynamic relations.

Additionally, this synthesis enables researchers to compare and contrast various patterns, such as the S-O, the S-R, the interplay, and the S-O-R patterns, to comprehend the contributions made by each to PR&PI research. For instance, encountering environmental stimuli does not guarantee that they will influence consumers' risk perception and affective state. In other cases, stimuli can directly trigger consumers' intentions. Similarly, factors in the organism, such as perceived risk, attitude, trust, emotion and satisfaction, are not correlated sequentially either. These findings contribute to confirming the validity of the overlapping SOR model.

Further, the review also contributes to explaining how variables come into play in the PR&PI relationship. For instance, the organism incorporates the experiential and memory factors through earlier encounters, making them distinct from affective states. Therefore, these factors can directly impact the consumer's cognitive and affective evaluations or combine with other stimuli at the time to change consumers' risk perception, affective state and purchase intention. This implies that consumers in e-commerce may change over time. Therefore, the relationships between stimuli, perceived risk, affect and intention are dynamic mechanisms.

7.2 | Practical implications

This review article also has significance for practitioners in e-commerce as well. First, the review presents two research streams on perceived risk, including uni- and multi-dimensions of perceived risk, which might be useful for the understanding of this concept and for developing several direct risk-reducing strategies. For instance, financial risk, time risk, functional risk and so on, may inspire the software to streamline the ordering process by storing the customer's data on the retailer's systems (Glover & Benbasat, 2010). Privacy risk, security risk, information risk and so on lead e-retailers and

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stakeholders to focus on providing better security in transactions and privacy protection for consumers' transactions (Chopdar et al., 2018).

Second, the SOR framework with various patterns provides an understanding of consumers' perceptions, emotions and responses in e-commerce. Therefore, e-retailers and e-commerce providers may recognize which stimulus factors influence consumers' risk perception and purchase intention. Consequently, this supports practitioners in promoting the effectiveness and efficiency of operations, marketing and communication strategies. The study also encourages the sustainability of these strategies by affirming that the mechanisms of stimuli affecting consumers' risk perception and purchase intentions are dynamic relations. Therefore, they may change over time.

8 | CONCLUSION

This study's premise recognizes that, despite consumers' familiarity with e-commerce and online shopping, they still prioritize issues related to perceived risk. The literature on the effect of perceived risk and purchase intention in e-commerce is rather fragmented, and the segmentation of the factors is still controversial. Therefore, this study conducts a systematic review to provide a holistic picture of the current status of this field. Research streams and modelling approaches to perceived risk in e-commerce are revealed. Further, the review also draws on the TCM and SOR frameworks to organize aspects and issues associated with PR&PI. The mechanism between perceived risk and purchase intention is not only cross-sectional but also dynamic. This effect also stems from various external and internal stimuli, affective states, and even experiential and memory factors of the consumer.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the Supporting Information of this article.

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